CHAPTER 1 INTRODUCTION

This chapter introduces the context of aircraft icing, its impact on flight safety, and its role in certification. It reviews the main ice morphologies, regulatory requirements, and certification tools, before presenting the challenges of numerical ice accretion modeling. The chapter concludes by defining the thesis objectives and outlining the organization of the thesis.

1.1 Context

In the context of climate change and increasingly stringent environmental regulations, the aeronautical industry is moving toward the development of more efficient and lower-emission aircraft. This transition introduces new aerodynamic, structural, and operational challenges, while maintaining flight safety as the primary requirement [8]. One by-product is the need for reliable certification procedures capable of assessing aircraft performance and safety over a wide range of operating conditions.

Among the phenomena considered during aircraft certification, in-flight icing presents significant challenges to aircraft safety and performance [9]. Ice accretion occurs when an aircraft flies through clouds containing supercooled water droplets[1]. When these droplets impinge on exposed surfaces and the local thermal conditions allow freezing, ice can accumulate on the aircraft geometry. The resulting ice shapes may significantly modify the aerodynamic configuration of the aircraft, leading to increased drag, reduced lift, altered stall behavior, and degraded controllability [9–12]. While aircraft icing is commonly associated with large exposed components such as wings, empennages, aircraft nose, and engines, it may also affect smaller but critical components, including probes and sensors, as illustrated in Figure 1.1. Ice accretion on these components can alter the aircraft response and produce erroneous measurements comprising flight safety.

Aircraft icing was not initially regarded as a primary concern during the early years of powered flight. Because aircraft were not yet equipped for routine instrument flight, pilots generally avoided clouds, and encounters with icing conditions were mostly inadvertent. This situation changed in the mid-1920s, when the U.S. Air Mail Service began operating scheduled day-and-night routes between New York and Chicago. These operations exposed pilots to icing conditions on a regular basis, and several serious incidents and accidents in 1926 and 1927 highlighted the need for practical solutions to the icing problem [9]. Early research

[1] Supercooled water droplets are liquid water droplets that remain unfrozen despite being at temperatures below the freezing point.



(a) Aircraft engine [13]



(b) Pitot probe [14]

Figure 1.1 Examples of ice accretion on aircraft components.

efforts were therefore mainly directed toward ice-protection systems (IPS)[2], including heated aircraft components, waterproof coatings, greased propellers, electrically heated pitot-static tubes, and pneumatic de-icing boots [9]. Although some of these solutions proved useful in specific applications, they remained largely empirical and did not provide a complete understanding of the physical mechanisms governing ice accretion.

Over the following decades, repeated claims that aircraft icing had been effectively solved proved premature, as operational experience continued to reveal unresolved challenges and the need for dedicated experimental investigation under controlled conditions. This need motivated the development of controlled icing research capabilities, culminating in the construction of the Icing Research Tunnel (IRT) in Cleveland in the 1940s [9]. The IRT enabled systematic investigations of supercooled cloud conditions, droplet impingement, aerodynamic performance degradation, and the performance of anti-icing and de-icing systems. These experimental studies played a central role in improving the physical understanding of aircraft icing and later provided essential data for the development, validation, and certification of numerical icing prediction methods [15]. These early efforts subsequently motivated the development of similar experimental icing facilities, which are discussed in Section 1.3.

1.2 Ice Morphologies

Depending on the atmospheric and aerodynamic conditions, different ice morphologies may develop on aircraft surfaces. At low temperatures, rime ice forms when supercooled droplets freeze almost immediately upon impact. This rapid solidification produces a relatively smooth, streamlined, and opaque ice shape. The opaque appearance of rime ice is asso-

[2]Ice protection systems are divided into two categories: anti-icing systems are designed to prevent ice formation on protected surfaces, whereas de-icing systems allow ice to form temporarily before removing it.

ciated with air trapped within the ice during the rapid freezing process.

As the temperature increases toward the freezing point, glaze ice may form. In this regime, the impinging droplets do not freeze completely upon impact. Instead, part of the liquid water remains on the surface, forming a thin water film that can run back before freezing downstream. Glaze ice is therefore usually denser and more transparent than rime ice and often leads to more complex ice shapes, including horn-like structures. On swept configurations, it can also produce 3D features such as ice feathers and larger-scale structures commonly referred to as scallops or lobster tails because of their characteristic resemblance to these forms (Fig. 1.2).

A third regime, mixed ice, occurs as a transition between rime and glaze conditions. In this case, glaze-like ice may form near the stagnation region, where the impingement rate is high and the water does not freeze completely on impact. Farther downstream, the remaining droplets or runback water may freeze more rapidly, producing rime-like deposits. Mixed ice therefore combines features of both rime and glaze ice.



Figure 1.2 Ice accretion on the leading edge of the Common Research Model (CRM) hybrid wing configuration tested in the NASA Icing Research Tunnel. Photos taken from [1].

1.3 Regulations & Certification

The diversity of ice morphologies and the severity of their aerodynamic effects make in-flight icing a critical consideration in aircraft certification. Regulatory authorities require aircraft to demonstrate safe operation within prescribed icing environments, which are defined through

envelopes of atmospheric and flight conditions. A major international reference for the certification of transport-category airplanes is Title 14 of the Code of Federal Regulations (CFR), Part 25, adopted by the Federal Aviation Administration (FAA). Within this framework, Appendix C [16] has historically defined the reference atmospheric icing envelopes used for the design and certification of IPS. These envelopes, in use since 1964, describe continuous maximum and intermittent maximum icing conditions in terms of parameters such as liquid water content, temperature, altitude, cloud horizontal extent, and representative droplet diameter [17]. They were primarily established for conventional cloud icing conditions, where the droplet median volume diameter (MVD) is typically limited to the range of standard supercooled droplets (SSD), up to approximately 50 μm.

However, several in-service events demonstrated that aircraft could also encounter icing environments outside the original Appendix C envelopes. A major example is the crash of American Eagle Flight 4184, an ATR-72, near Roselawn, Indiana, on October 31, 1994 [18]. The accident investigation identified icing as a central factor in the loss of control and highlighted the safety implications of supercooled large droplet (SLD) conditions, including freezing drizzle and freezing rain. These environments may contain droplets significantly larger than those considered in Appendix C, with droplet diameters reaching several hundred micrometres, typically up to about 500 μm in freezing-drizzle conditions, and up to approximately 1000 μm in freezing-rain conditions.

This recognition led to additional regulatory developments, culminating in the introduction of Appendix O for SLD icing conditions [19]. Appendix O extends the certification envelope to include freezing drizzle and freezing rain, thereby addressing icing scenarios in which large droplets may splash, rebound, or break up before freezing. These mechanisms may lead to ice formation downstream of the protected region. Consequently, aircraft seeking certification for flight in icing conditions must demonstrate safe operation under these conditions.

The certification of IPS may rely on several complementary approaches, including flight testing, icing wind-tunnel testing, and numerical simulation [20]. Flight tests provide the direct assessment of aircraft behavior in realistic operating conditions and may be conducted either in natural icing environments or using artificial ice shapes. However, such tests are expensive, difficult to reproduce, and may involve safety risks, particularly when degraded aerodynamic performance or controllability must be evaluated.

Icing wind tunnels therefore became an essential complementary experimental tool for studying ice accretion and evaluating IPS performance under controlled conditions. Facilities such as the NASA IRT, the National Research Council (NRC) Low-Speed Icing Wind Tunnel in Canada [21], the Rail Tec Arsenal (RTA) facility in Austria [22], and the Office National

d'Études et de Recherches Aérospatiales (ONERA) icing facilities [23] in France have contributed extensively to the development of experimental databases for icing research and certification support. Compared with flight testing, wind-tunnel experiments offer better control of atmospheric parameters, and detailed access to local measurements. Nevertheless, they remain costly and face important scaling limitations, especially as aircraft components increase in size. Reproducing full-scale icing conditions often requires preserving several competing similarity parameters, including the relationships between droplet size, leading-edge radius, liquid water content, velocity, temperature, and accretion time. In practice, the model geometry or operating conditions must often be adjusted to fit within the facility constraints, and these adjustments can themselves require dedicated extensive analyses [24].

In addition, ice accretion experiments are subject to variability from one test to another, owing to uncertainties in cloud calibration, droplet distributions, thermal conditions, surface roughness, and measurement procedures. These limitations have motivated the development of numerical icing tools as a complementary approach to flight and wind-tunnel testing.

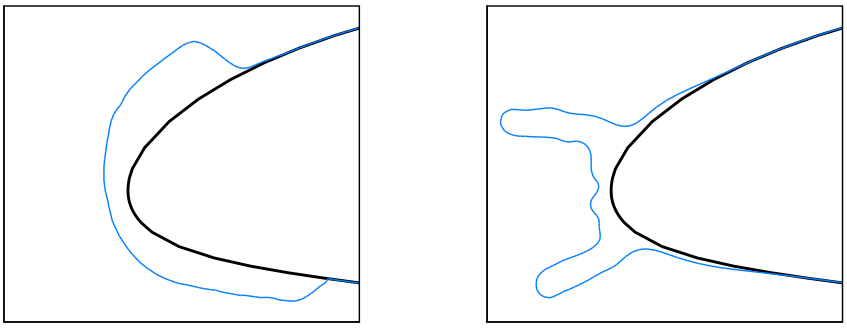
1.4 Ice Accretion Numerical Modeling

Ice accretion results from the interaction of several physical processes, including the aerodynamic flow around the aircraft, the transport and impingement of supercooled water droplets, the local thermodynamic balance at the surface, and the progressive modification of the geometry due to ice growth.

Although ice accretion is an unsteady phenomenon, most computational icing solvers rely on a quasi-steady approximation. This assumption is justified by the separation between the short characteristic time scales of the airflow and droplet motion and the much longer time scale associated with ice growth. In practice, the total exposure time is divided into one or several discrete accretion steps. Within each step, all the physical processes are simulated under steady-state conditions [25].

The simplest implementation consists of performing the entire accretion process in a single step. This approach is attractive because it limits the computational cost and avoids repeated geometry and mesh updates. However, it assumes that the flow, droplet trajectories, and surface thermodynamic solutions remain representative throughout the full exposure time, which can reduce the accuracy of the predicted ice shape. A more representative strategy consists of using a multilayer approach, in which the geometry is updated several times during the accretion process and the physical fields are recomputed on the evolving surface. The influence of these successive updates on the predicted ice shape is illustrated in

Fig. 1.3. Consequently, the computational cost increases with the number of accretion steps, as the aerodynamic, droplet-impingement, and thermodynamic solutions must be repeatedly recomputed on the evolving geometry. For 3D configurations, the size of the computational meshes and the repeated solution of these coupled physical models can result in substantial computational requirements.



(a) Single-step ice shape. (b) Multilayer ice shape.

Figure 1.3 Comparison of single-step and multilayer ice accretion predictions on a NACA 23012 airfoil at iso-physical parameters.

By accounting for the progressive modification of the geometry, the multilayer approach provides a stronger coupling between ice growth and the surrounding physical fields. This improvement comes at the cost of additional computations and repeated geometry and mesh updates, which introduce further computational robustness and accuracy challenges. In particular, successive geometry deformation or remeshing may progressively deteriorate the mesh quality, leading to highly skewed, tangled, or invalid elements. This degradation in mesh quality can adversely affect the numerical solvers by increasing numerical dissipation and stiffness, thereby reducing solution accuracy and convergence performance. In more severe cases, highly distorted or invalid elements may lead to solver instability and ultimately numerical failure. Such deterioration of the mesh elements is illustrated in Fig. 1.4 on a 30° swept NACA 0012 wing.

1.5 Problem Statement

Two-dimensional icing simulations have played a central role in the development and validation of ice accretion models. They remain computationally efficient and are particularly useful for airfoil configurations, parametric studies, and comparisons with well-controlled experimental databases. In some applications, they are also extended to approximate 3D effects

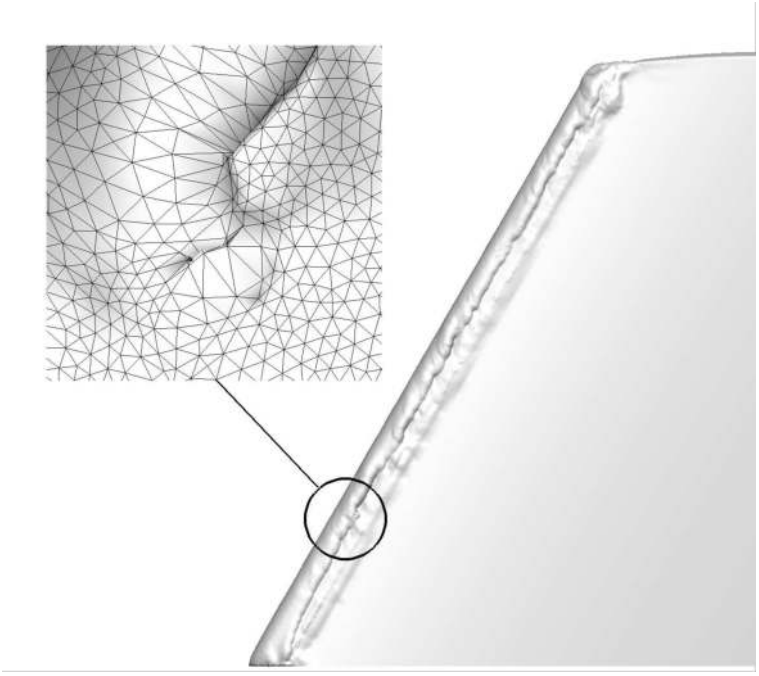


Figure 1.4 Mesh entanglement on a 30° swept NACA 0012 wing during multilayer ice accretion.

through 2.5D approaches [26]. However, such methods remain limited when the ice accretion process is governed by genuinely 3D flow features, spanwise transport, complex geometry effects, or non-uniform impingement patterns.

Fully 3D simulations provide a more general framework for predicting ice accretion on realistic aircraft components and complex configurations. Nevertheless, they are significantly more demanding in terms of computational resources and numerical robustness. The large computational meshes required to resolve 3D geometries, combined with the repeated solution of the aerodynamic, droplet, and thermodynamic models in a multilayer framework, make high-performance computing (HPC) an important component of practical 3D ice accretion simulations. Efficient parallel algorithms and scalable numerical methods are therefore required to maintain tractable computational times as the problem size and number of accretion steps increase. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. These difficulties become particularly important in multilayer simulations, where the geometry must be updated repeatedly as the ice grows. Each update may degrade the mesh quality or generate invalid surface and volume elements (Fig. 1.4). As a result, a simulation that has already required substantial computational effort may fail before reaching the final accretion time.

This robustness constraint partly explains why single-step ice accretion strategies remain

widely used in practical applications. They are simpler to execute and reduce the risk of mesh failure, but they also limit the ability of the solver to account for the feedback between the evolving ice shape and the surrounding flow and droplet fields. In the Ice Prediction Workshop (IPW)[3], many submitted ice-shape predictions relied on such single-step approaches. While effective from a computational standpoint, these methods generally showed reduced ability to reproduce complex accretion features, especially when compared with multilayer simulations [27,28]. This highlights the need for numerical strategies capable of preserving the robustness of single-step approaches while retaining the improved physical fidelity of multilayer ice-growth modeling.

1.5.1 Motivation for the Present Work

The preceding discussion identifies the repeated update of the iced geometry as a major limitation of 3D multilayer ice accretion simulations. Although the flow, droplet, and thermodynamic solutions must be recomputed consistently as the ice shape evolves, the ability to reach the final accretion time often depends on the robustness of the geometry-update procedure itself.

This motivates the use of geometry representations that are less sensitive to local surface irregularities. Previous studies [2, 29, 30] have demonstrated the potential of level-set descriptions for representing and evolving iced geometries in 2D configurations. In a level-set formulation, the interface is not tracked explicitly by the surface mesh. Instead, it is represented implicitly as the zero level of a scalar function ϕ. The sign of ϕ distinguishes the ice and air regions, while the interface is recovered from the condition $\phi = 0$.

Building on these advances, the present work extends this idea to 3D multilayer ice accretion simulations, where geometry updates become more challenging due to spanwise variations and increasingly complex ice features. By representing the iced surface implicitly, the geometry can be displaced and reconstructed in a more systematic manner, while preserving the clean portions of the geometry that are not modified by ice accretion. This reconstruction strategy reduces the sensitivity of the workflow to local geometric defects and provides greater control over the quality of the reconstructed surface mesh.

Since geometry evolution is only one component of the icing simulation framework, the preceding physical models must also provide reliable inputs to the accretion process. In particular, droplet impingement predictions determine the local water mass flux supplied to the thermodynamic model and directly influence the ice-growth rate. For this reason,

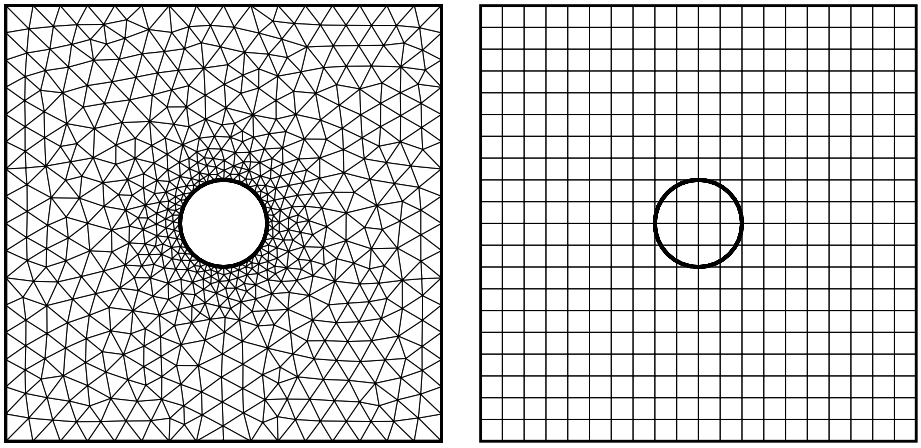
[3]The Ice Prediction Workshops (IPW) [27,28] are an AIAA international benchmark initiative for assessing state-of-the-art in numerical aircraft icing tools against experimental data.

Eulerian and Lagrangian droplet formulations, including their extension to Supercooled Large Droplet conditions, are also examined in this thesis.

Finally, to reduce the mesh-generation cost associated with successive geometry updates in an icing framework, growing interest has been directed toward immersed-boundary methods (IBMs) [30–32].

In a body-fitted approach, the computational mesh conforms to the solid surface, as shown in Figure 1.5a. As the ice geometry evolves, maintaining this conformity generally requires deformation or regeneration of the surrounding volume mesh. In contrast, an immersed-boundary method embeds the geometry within a non-conformal background mesh, as shown in Figure 1.5b. The geometry is therefore represented independently of the mesh, which can simplify mesh generation and geometry handling for complex or evolving iced surfaces.

This decoupling between the geometry and the computational mesh makes immersed-boundary methods particularly attractive for multilayer ice accretion, where the surface evolves repeatedly throughout the simulation. In this context, their integration within a complete icing framework provides a potential means of reducing the meshing constraints associated with conventional body-fitted approaches.



(a) Body-fitted mesh. (b) Immersed-boundary mesh.

Figure 1.5 Mesh representation around a cylinder using (a) a body-fitted mesh and (b) an immersed-boundary mesh.

1.5.2 Objectives of the Present Study

The primary objective of this thesis is to improve the robustness of 3D multilayer ice accretion simulations through level-set-based geometry evolution. The proposed developments aim to

reduce geometric failures during the accretion process, improve the quality of reconstructed iced surfaces, and maintain reliable aerodynamic, droplet-impingement, thermodynamic, and ice-shape predictions throughout the icing simulation framework.

More specifically, this work develops and assesses numerical methods required to support robust multilayer ice accretion simulations, including Lagrangian particle tracking for droplet impingement prediction, Supercooled Large Droplet effects in Eulerian and Lagrangian formulations, level-set-based geometry update and surface reconstruction, and an immersed-boundary extension of the existing icing framework.

To accomplish this objective, the following sub-objectives are defined.

O.1: Develop, Accelerate, and Assess a Lagrangian Droplet Impingement Model and Extend Droplet Impingement Modeling to Supercooled Large Droplet Conditions

The first objective is to develop and evaluate Lagrangian particle-tracking methods for predicting droplet trajectories and collection efficiency.

The method is assessed in terms of impingement limits, collection-efficiency distributions, computational cost, and parallel performance, and is compared with the existing Eulerian droplet model.

This objective also extends both the Eulerian and Lagrangian impingement models to account for Supercooled Large Droplet effects. These developments are used to assess the influence of large-droplet effects on droplet collection and impingement limits.

O.2: Develop and Assess a Robust Level-Set Methodology for 3D Multilayer Ice-Shape Evolution

The second objective is to develop a robust methodology for multilayer ice-shape evolution using a level-set description of the evolving geometry. The method is intended to support geometry displacement, surface reconstruction, and preservation of the clean portions of the geometry, while reducing geometric failures during multilayer simulations.

The methodology is then applied and assessed on representative 2D and 3D icing configurations. These cases are used to demonstrate its applicability to multilayer ice accretion and to examine its behavior for increasingly complex geometries.

O.3: Develop, Integrate, and Assess an Immersed-Boundary Framework for Ice Accretion Simulations

The third objective is to extend an existing immersed-boundary flow solver by integrating Eulerian and Lagrangian droplet impingement, thermodynamic ice accretion, and geometry-evolution capabilities within a unified ice-accretion framework. The objective is to demonstrate the feasibility of performing the complete ice-accretion sequence without repeated body-fitted volume remeshing and to establish a numerical foundation for the further development of immersed-boundary icing methods.

The resulting immersed-boundary framework is then assessed against established body-fitted formulations using selected verification and validation cases. Comparisons of droplet collection efficiencies and final ice shapes are used to evaluate the agreement between the two approaches and identify areas requiring further development.

1.6 Thesis Outline

This thesis is organized in a hybrid format, combining conventional chapters with one article-based chapter. Chapter 2 first presents the state of the art in numerical ice accretion modeling. It reviews the main components of icing simulation workflows, including aerodynamics, droplet impingement, SLD effects, thermodynamics, geometry evolution strategies, and immersed-boundary methods.

Chapter 3 focuses on Lagrangian particle tracking for droplet-impingement prediction under Supercooled Large Droplet conditions. Particular attention is given to the treatment of large-droplet effects and the accuracy of the predicted collection-efficiency distributions, as well as the computational efficiency and parallel scalability of the method for large-scale simulations.

Chapter 4, presented as the submitted journal article, addresses the problem of robust geometry evolution in 3D multilayer ice accretion simulations. A level-set-based approach is developed and assessed as a means of updating complex ice shapes while reducing the risk of mesh degradation and invalid geometries during the accretion process.

Chapter 5 extends the developments introduced in the preceding chapters to an immersed-boundary framework for ice-accretion simulations. This chapter investigates the use of immersed-boundary methods as an alternative to body-fitted approaches for aerodynamic and droplet-impingement calculations.

Chapter 6 provides a general discussion comparing the main numerical methodologies investigated throughout the thesis, including Eulerian and Lagrangian approaches for droplet

impingement, algebraic and level-set methods for geometry evolution, and body-fitted and immersed-boundary approaches for ice-accretion simulations. Finally, Chapter 7 presents a synthesis of the main contributions, followed by the limitations of the proposed methods and recommendations for future work.